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# EXERCISE SHEET / THEME 6: MEASURES OF DISPERSION.

## EXERCISE 1. SUBSIDIES AND TAXES, SHIFTING AND SCALING.

### First part. Original setting.

You receive the following data set with 5 observations. They represent the income of five persons.

| | | | | |
|---|---|---|---|---|
| 100 | 110 | 120 | 450 | 900 |

Estimate the following statistical measures for the given data set:

a) Mean
b) Percentile 25
c) Percentile 50 (or median)
d) Percentile 75
e) Variance
f) Standard deviation
g) Interquartile range (IQR)

### Second part. Effect of shifting the data.

You hear that the government is about to start a universal subsidy of 20 units. The following data set represents this income increase. You want to analyze what will be the situation of your sample, so you increased the income by 20 units.

| | | | | |
|---|---|---|---|---|
| 120 | 130 | 140 | 470 | 920 |

h) Repeat the calculations from the first part (from a to g).

### Third part. Effect of scaling the data.

Finally, you also know that the government is planning to create a flat tax of 10%. The following data set represents the final income considering the subsidy but also the 10% tax.

| | | | | |
|---|---|---|---|---|
| 108 | 117 | 126 | 423 | 828 |

i) Repeat the calculations from the first part (from a to g).

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**EXERCISE 2. SUMMARY TABLE.**

a) Summarize all the statistical measures for each of the three scenarios in a final table.
b) Which are central tendency measures?
c) What happens to them when we move from the first to the second scenario (shifting the data)? What about when moving from the second to the third scenario (scaling the data)?
d) Which are spread measures (or dispersion measures)?
e) What happens to them when we move from the first to the second scenario (shifting the data)? What about when moving from the second to the third scenario (scaling the data)?

**EXERCISE 3. BOX PLOTS.**

a) Draw three box plots, one for each of the three different scenarios. Hint: the box is defined by Q1 (lower limit), Q2 (or median as the middle line), and Q3 (upper limit). Extend the whiskers so that the lower whisker reaches the minimum observation and the upper whisker reaches the maximum observation.